Neighbors Help You Best: Geographical Adjacent Perception for Remote Sensing Long-Tailed Semantic Segmentation

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***Abstract*—While semantic segmentation has witnessed remarkable advancements in remote sensing (RS), the deployment of these models is severely hampered by the inherent long-tailed distribution. Despite extensive progress in classification and object detection tasks, long-tailed learning in pixel-wise segmentation remains critically underexplored. Major challenges include the lack of dedicated datasets, ambiguity in defining long-tailed classes, and the inability of generic methods to leverage intrinsic geographic relationships for bias mitigation. To bridge these gaps, we first construct the GaoFen multiclass semantic segmentation dataset (GMSD), a large-scale long-tailed semantic segmentation dataset comprising 21 fine-grained categories with a spatial resolution of 0.8 m. Then, a critical threshold is identified to define head and tail classes based on empirical performance variations in different proportions, thus providing a unified foundation for balanced learning and consistent evaluation. Finally, we propose the geographical adjacent perception (GeoAP)-based framework, which leverages boundary-dependent relationships between land cover types to achieve robust predictions. Specifically, GeoAP first explicitly models interclass contextual adjacency and incorporates a dynamic tail sample container, expanding tail-category distribution coverage while preserving geographical plausibility. After that, it further captures spatial adjacency intensity to guide unbiased learning and suppress implausible predictions. Based on our partition strategy, extensive experiments on GMSD, GID, and FBP datasets demonstrate the effectiveness and superiority of the proposed GeoAP. The code and dataset will be released at https://github.com/skyflyi/GeoAP-Segmentation**

***Index Terms*—Adjacent perception, datasets, long-tailed learning, remote sensing (RS), semantic segmentation.**

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I. Introduction

SEMANTIC segmentation of remote sensing (RS) serves as a fundamental task in earth observation [1], [2], attracting significant attention across the computer vision and geoscience communities. By assigning pixel-level semantic labels, this technology underpins a myriad of critical applications, including precision agriculture, urban planning, disaster assessment, and so on [3], [4].

However, the real-world scenarios inherently exhibit a long-tailed distribution in the spatial and statistical characteristics [5], [6]. Specifically, a small set of dominant "head" categories occupies the majority of pixels, whereas the remaining numerous "tail" categories account for only a small proportion [7]. This severe imbalance hinders the model's ability to effectively extract and discriminate features for tail classes, leading to a significant drop in segmentation generalization across complex scenes [8]. Compounded by intrinsic challenges including blurred boundaries, multiscale confusion, and background noise [9], the long-tailed semantic segmentation remains largely under-explored. In this work, we identify three fundamental barriers that impede its progress: the lack of benchmark datasets, the ambiguous definition of head–tail classes, and the neglect of RS geospatial-specific information in existing methods.

First, due to the high costs of annotation, commonly used semantic segmentation benchmarks, such as LoveDA [10] and ISPRS Potsdam [11], offer relatively accurate annotations but contain only a small number of categories, typically only six and seven classes. While other fine-grained datasets, including GID [12], provide rather broader coverage and rich semantic labels, they are limited by a low spatial resolution, high background ratios, and poor image quality. As a result, high-quality benchmarks dedicated to long-tailed RS semantic segmentation remain insufficient. Second, unlike classification and detection tasks, where head and tail classes are clearly separated by instance counts, semantic segmentation inherently involves dense pixel-level prediction and spatial dependencies. This makes it difficult to define a reasonable and consistent criterion for head–tail class partitioning, hindering reliable performance evaluation and effective optimization. Third, most existing long-tailed learning strategies are developed from generic statistical perspectives, such as sampling [13], [14] or loss re-weighting [15], while the intrinsic geographical relationships among land-cover categories

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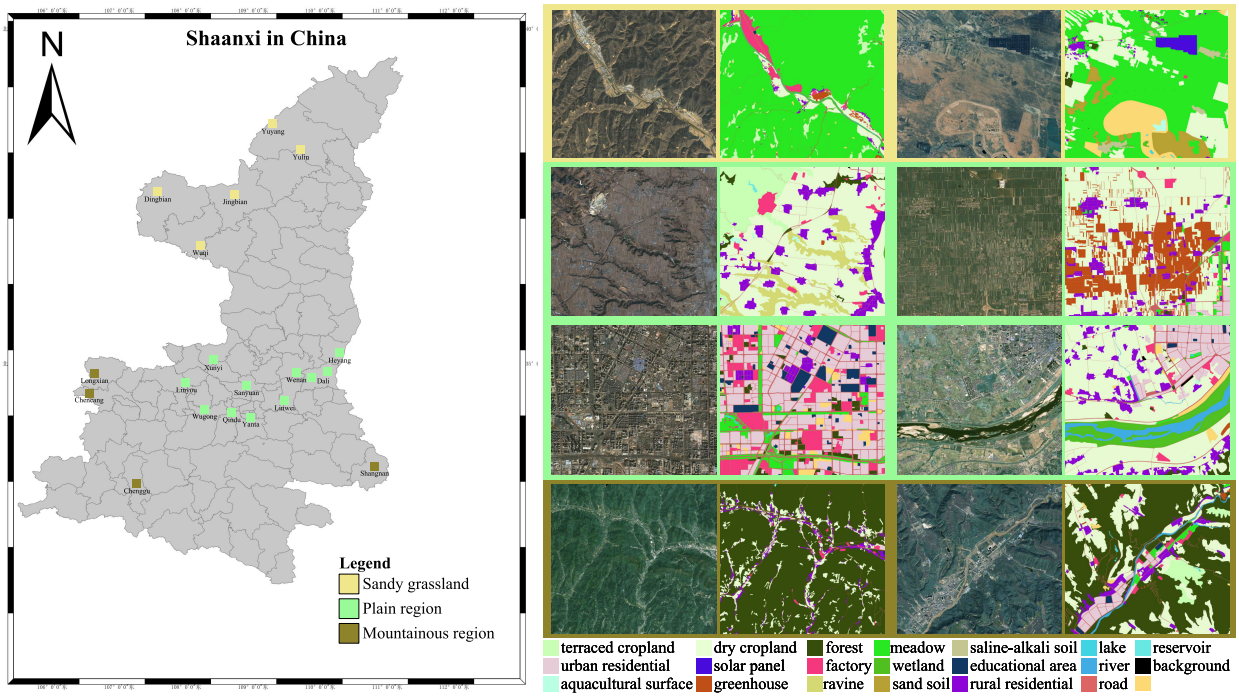


Fig. 1. Geographic distribution of the GMSD dataset. The left panel illustrates the spatial distribution across Shaanxi Province in China, covering representative landscapes including sandy grassland, plain region, and mountainous region. The right panels present representative RS images and their corresponding annotations.

are rarely exploited. In practice, land-cover categories often exhibit structured spatial adjacency rather than independent distributions, which can provide valuable prior knowledge to improve tail-class learning. This oversight restricts the model's generalization ability across diverse terrains and scenarios, making it difficult to achieve stable and reliable segmentation results.

To address these issues, we propose a systematic framework for RS long-tailed semantic segmentation. A comprehensive benchmark dataset named GMSD is first constructed as shown in Fig. 1. It encompasses numerous fine-grained semantic categories and spans diverse geographical regions, including mountainous terrain, plains, and sandy grasslands. Moreover, motivated by the work of Anand et al. [16] analyzing long-tailed learning differences in a simple binary-class problem, we conduct comprehensive binary segmentation experiments to find a reasonable way of grouping for evaluation.

Finally, we propose a geographical adjacent perception (GeoAP)-based framework, which explicitly exploits the geographical adjacent dependency to promote balanced learning. Specifically, GeoAP consists of two key components: the adjacency-filtered tail augmentation (AFTA) module and the adjacent-intensity perception (AIP) loss. By synthesizing tail-class regions within their semantically valid neighbors, AFTA effectively broadens the contextual scene distribution while preserving geographical consistency, which in turn enhances the model's learning bias toward tail categories. AIP is introduced to guide the network in further learning the geographical adjacency degree across different classes. This approach reduces misclassification of geographically mutually exclusive categories and facilitates learning adjacent information of the tail classes. This dual strategy of GeoAP not only yields remarkable improvements for tail classes but also preserves performance in head classes, with consistent transferability across diverse network architectures and datasets.

In summary, the main contributions of our work are as follows.

1) We present GMSD, a novel multiclass RS segmentation dataset characterized by imbalanced distributions and precise annotations, which serves as a valuable benchmark for future work in addressing long-tailed challenges.
2) We introduce a reliable grouping criterion for defining head and tail classes, providing a foundation for consistent evaluation and targeted optimization in subsequent long-tailed research.
3) We propose GeoAP, which leverages geographic adjacency among ground objects to facilitate balanced learning for long-tailed semantic segmentation. Within this framework, AFTA ensures geographically plausible augmentation to enhance the learning preference for tail classes, while AIP utilizes adjacency strength information to mitigate class confusion.

The rest of this article is organized as follows. Section II describes the existing related works. Section III introduces the details of the proposed GMSD dataset, the criteria for grouping head and tail classes, and the GeoAP method. Experiments are demonstrated in Section IV. Finally, we draw the conclusions in Section V.

II. Related Work

A. Datasets for Long-Tailed Semantic Segmentation

The performance of long-tail semantic segmentation algorithms is highly dependent on the quality and rationality of the

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TABLE I
COMPARISON OF GMSD WITH CURRENT MAIN SEMANTIC SEGMENTATION DATASETS IN RS

Dataset	Classes	Annotation granularity	Resolution(m)	Image quality	Background coverage(%)	Area(km^2)	Year
ISPRS Potsdam [11]	6	Coarse	0.05	4.43	28.70	3.42	2013
ISPRS Vaihingen [17]	6	Coarse	0.05	5.67	27.81	1.38	2013
WHDLD [18]	6	Coarse	2.0	5.50	0.00	1294	2018
DeepGlobe 2018 [19]	7	Coarse	0.5	7.10	0.05	1716	2018
LoveDA [10]	7	Coarse	0.3	5.79	3.31	536.15	2021
URUR [20]	8	Coarse	1.0	5.46	0.14	78852	2023
GID* **[12]**	16	Fine	4.0	10.56	37.47	313	2020
FBP [21]	25	Fine	4.0	9.57	32.09	7555	2023
GMSD(ours)	21	Fine	0.8	10.93	0.05	776	2026

"" denotes the Fine Land-cover Classification set of GID. Annotation granularity is set to coarse when classes is less than 10. Image quality is the images clarity assessed by the average gradient. Background coverage denotes the proportion of background category. Area indicates the land cover extent of the entire dataset.*

dataset design. Existing RS semantic segmentation datasets have promoted the rapid progress of this field, but many are not specifically designed for long-tailed segmentation. As shown in Table I, we define annotations with fewer than ten classes as coarse-grained dataset, providing only a preliminary annotation of ground objects. It can be observed that commonly used benchmarks, including ISPRS Potsdam and ISPRS Vaihingen [17], contain only six typical land cover classes, mainly focusing on several typical urban land-cover categories. Subsequent datasets like LoveDA and URUR [20] expanded coverage and category counts, yet still predominantly featured coarse-grained annotations of seven and eight categories, focusing on macro-level feature recognition. This limited semantic granularity fails to capture the long-tailed distribution inherent in RS scenarios, thereby hindering refined downstream applications such as photovoltaic coverage assessment and urban–rural zoning.

Fine-grained datasets like GID [12] and FBP [21] provide richer category annotations and are more suitable for studying long-tailed segmentation. However, they were not originally designed for long-tailed issues, making them not fully adapted to this area. First, their spatial resolution is limited to 4 m, far below the sub-meter requirements for practical applications. At this resolution, small objects like solar panels and aquacultural surfaces suffer from pixel overlap and shape loss, causing severe degradation of key features such as texture and contours. Moreover, GID has limited geographic coverage, containing only ten regions dominated by plains cities, lacking typical scenes like mountains and rural areas. FBP suffers from texture and spectral information loss in some images, which leads to low image quality and partially distorted building outlines. Inconsistent image quality will introduce more annotation noise, disrupting long-tailed learning. Therefore, we use the average gradient metric to evaluate image definition. The results in Table I demonstrate that FBP performs relatively poorly in fine-grained datasets.

In addition, existing datasets typically exhibit disproportionately high background ratios. We report the background proportion for each dataset in Table I, among which the GID dataset features a ratio as high as 37.47%. For long-tailed categories with inherently sparse samples, the feature interference of the background may hinder the model's ability to focus on learning rare object features. This not only increases the model's computational burden but also exacerbates classification bias due to insufficient samples, ultimately compromising segmentation accuracy and generalization capabilities in long-tailed learning.

To alleviate these limitations, this article constructs GMSD, a high-resolution, fine-grained RS semantic segmentation dataset. Compared with existing datasets, GMSD provides 21 semantic categories, 0.8 m spatial resolution, low background coverage, and diverse geographical scenarios. Therefore, it can better support the evaluation of long-tailed semantic segmentation methods under realistic RS conditions.

B. Long-Tailed Learning

Long-tailed learning has been extensively explored in image classification [22] and object detection tasks [23], where samples appear as discrete instances. This characteristic allows for straightforward head–tail partition based on sample counts: head (more than 100 instances), middle (20–100 instances), and tail (less than 100 instances).

However, semantic segmentation is rather more difficult to group, characterized by its dense pixel-wise nature. Cui et al. [24] adopt a rather vague grouping strategy in research, while Zermatten et al. [25] applied a division based on personal intuition. Li et al. [26] followed a fixed ratio inspired by classification tasks. All these approaches suffer from a lack of adequate theoretical and experimental backing, leading to disparate evaluations. Therefore, a reliable head–tail grouping criterion is necessary for fair comparison and targeted optimization in long-tailed semantic segmentation.

In addition, to mitigate the learning bias caused by long-tailed distributions, existing methods can be roughly divided into data-level, loss-level, and representation-level strategies.

Data re-balancing methods adjust the training distribution by oversampling tail classes [13], [27], undersampling head classes [14], [28], or generating additional samples through data augmentation [29], [30]. These strategies can reduce the dominance of head categories, but they are less straightforward in semantic segmentation because pixel-level labels must preserve spatial continuity and contextual consistency. Directly copying or resampling rare regions may destroy the geographical structure of RS scenes and introduce unrealistic training samples.

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Loss-based methods modify the optimization objective to emphasize difficult or underrepresented categories. Focal Loss [15] reduces the contribution of easy samples, Dice Loss [31] improves region-level optimization, and cost-sensitive losses [32], [33] assign category-dependent weights according to imbalance ratios. CFL [34] further enhances tail learning by systematically incorporating prediction difficulty into the optimization objective. Although these methods can strengthen tail-class learning to some extent, excessive re-weighting may lead to overfitting on rare categories or even degrade the performance of head classes.

Recent studies have also explored architectural and feature-level strategies for long-tailed semantic segmentation. Wang et al. [35] proposed BLV to enhance the separability of tail classes in the feature space, while Aucseg [36] designed a novel optimization function based on the area under the ROC curve to balance head and tail learning. Li et al. [26] introduced dynamic matcher adjustment within Mask2Former to achieve balanced representation learning. Zermatten et al. [25] developed MCE, which trains multiple expert networks separately for head and tail classes before combining their outputs, but increases computational complexity and sacrifices head class precision. SMRF [37] employed multiscale data augmentation and multimodal fusion to alleviate imbalance, yet its focus is primarily on ultrahigh-resolution (UHR) satellite imagery and overall metrics, with limited emphasis on tail class improvement.

However, these methods mainly address long-tailed learning from generic statistical or architectural perspectives, while the geographical characteristics of RS imagery are still insufficiently explored. Different from these methods, this work focuses on the unique spatial regularity of RS scenes. Instead of only adjusting sample frequency or loss weights, GeoAP explicitly models geographical adjacency relationships among land-cover categories to achieve more balanced segmentation.

C. Domain-Aware RS Object Understanding and Geographic Relationship Modeling

Recent RS studies have explored various domain-aware strategies for difficult target interpretation, including low-prevalence categories, small objects, and feature matching. These works are related to long-tailed semantic segmentation because they also attempt to improve the recognition of challenging targets under sparse appearance, weak structure, or heterogeneous imaging conditions.

Foundation-model-based methods have shown promising potential in RS object understanding. ROS-SAM [38] adapts SAM to high-quality interactive segmentation of RS moving objects, showing the potential of foundation models for sparse or low-prevalence target segmentation. Xie et al. [39] further explored SAM-based slender object segmentation by introducing learnable morphological skeletons to preserve structural continuity and topological completeness. However, they generally focus on interactive segmentation or morphology modeling rather than category-level long-tailed learning.

Multimodal and multisource methods provide another perspective. Chen et al. [40] proposed a frequency-domain fusion strategy for multimodal RS semantic segmentation, aiming to exploit complementary spectral and structural information from heterogeneous modalities. MARSNet [41] introduced a Mamba-driven adaptive framework for robust multisource RS image matching under noisy conditions. These methods improve feature representation or cross-source correspondence, but their primary objectives are not to mitigate the optimization bias between head and tail categories.

These studies indicate that remote-sensing-specific priors are important for difficult target understanding. However, existing approaches mainly focus on interactive segmentation, morphological structure preservation, multimodal feature fusion, or multisource correspondence. They do not explicitly exploit geographical adjacency relationships to guide long-tailed semantic segmentation. In contrast, GeoAP is designed for fully automatic category-level long-tailed segmentation. It introduces geographical adjacency priors into both sample augmentation and loss optimization. Specifically, the AFTA module expands tail-class contextual distributions while preserving geographical plausibility, and the AIP loss suppresses unreasonable predictions between geographically incompatible categories. In this way, GeoAP provides a domain-aware and complementary solution for RS long-tailed semantic segmentation.

III. Proposed Method

In this section, we first provide a detailed introduction to the annotation process of our GMSD dataset in Section III-A. Subsequently, in Section III-B, we demonstrate the criterion for grouping head and tail classes in long-tailed semantic segmentation. After that, in Section III-C, we detail the AFTA module, which performs data pruning and enhancement for tail classes based on the geographical adjacency information of ground objects. Finally, in Section III-D, we introduce the AIP loss, designed to further enhance the discrimination capability for tail categories.

A. GMSD Dataset

To reduce the gap between long-tailed datasets in semantic segmentation and real-world application requirements, the GMSD dataset is constructed as shown in Fig. 1. It contains diverse geographic coverage and 21 fine-grained semantic categories, which cover: terraced cropland, urban residential, aquacultural surface, dry cropland, solar panel, greenhouse, forest, factory, ravine, meadow, wetland, sand soil, saline-alkali soil, educational area, rural residential, lake, river, road, reservoir, bare surface, and background. This comprehensive system is designed to broaden geographical analysis scenarios by including both commonly seen objects and distinctive categories (e.g., solar panel, aquacultural surface), thereby accurately reconstructing real-world distribution patterns.

We use GF-7 satellite imagery as the data source for the GMSD dataset. GF-7 satellite is the first civil sub-meter stereo mapping platform launched by the Ministry of China Natural Resources, featuring a dual-line array stereo camera with a 20-km swath width. It acquires 0.8-m forward/backward panchromatic (Pan) and 3.2-m multispectral (MS) imagery. Existing datasets, including GID and FBP, typically rely on

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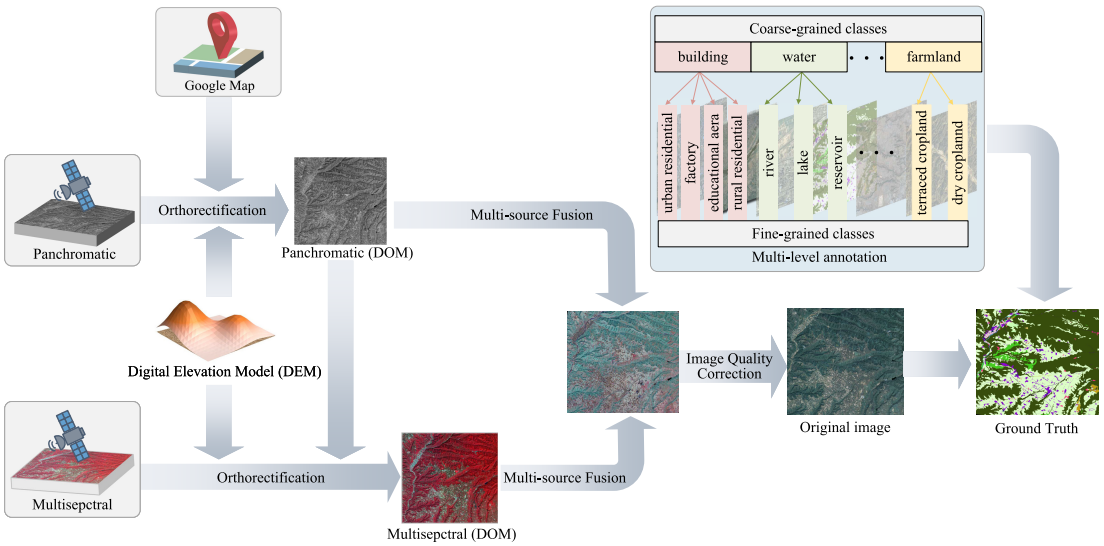


Fig. 2. Data processing of high-resolution images obtained through the fusion of multispectral and panchromatic data.

4-m MS imagery, compromising spatial fidelity. In contrast, the GMSD dataset carries a multisource fusion scheme that integrates 3.2-m MS data with 0.8-m backward Pan imagery, which retains spatial details while preserving authentic spectral (RGB) information.

Fig. 2 demonstrates the pipeline used to generate the final 0.8-m annotated images for the GMSD dataset from GF-7 raw data. To ensure geometric precision and consistency, the fusion processing is implemented based on Geomatica. First, the backward Pan imagery undergoes orthorectification referencing a digital elevation model (DEM) and Google Maps. This step eliminates projection errors and geometric distortions, generating high-quality digital orthophotography maps (DOMs) with precise geographical location. Subsequently, the MS imagery is co-registered and orthorectified using the processed Pan imagery and DEM data. This synchronous correction ensures strict spatial alignment between modalities, mitigating pixel misalignment caused by subtle variations in shooting angles. After that, the Pansharping algorithm fuses MS and Pan data to produce high-resolution images. Professional color correction is finally performed using Photoshop to harmonize visual consistency with ground truth features, producing high-quality images. Through this meticulous processing, the GMSD dataset achieves a synergy of 0.8-m GSD and true-color spectral fidelity.

The creation of the GMSD dataset is fully a manual annotation paradigm, executed by geoscience experts and subject to a "norm-guided, hierarchically refined, and iteratively verified" protocol. Adhering to the Technical Regulation for the Third Chinese National Land Survey (TD/T 1055-2019), we implement a top-down hierarchical annotation. Coarse-grained primary categories (e.g., buildings) are subdivided into fine-grained classes (e.g., rural residential, factory, urban residential). This hierarchy significantly enhances semantic richness, facilitating multilevel feature extraction. Annotations are performed using ArcGIS, where boundaries are delineated with pixel-level precision to ensure alignment with feature patches. After multiple rounds of assessment on classification accuracy and boundary integrity, we obtained the GMSD dataset with low label noise and high quality.

To ensure geographic representativeness, samples were curated from three geologically distinct regions in Shaanxi Province: Sandy Grassland in Northern Shaanxi, the Plain region in Guanzhong, and the Mountainous region in Southern Shaanxi. These regions encompass diverse land covers, ranging from complex humanistic environments to unique natural landscapes. Furthermore, multiseasonal imagery is incorporated to enhance temporal diversity. As shown in Fig. 1, the dataset consists of 20 samples, each with a size of 7400×8200, covering approximately 776 km^2 to ensure robust semantic variety.

A common limitation of existing fine-grained datasets (e.g., GID and FBP) is the predominance of background pixels, which may dominate pixel statistics and dilute effective supervision for rare categories. Instead, GMSD reduces the proportion of background pixels to 0.05% in Table I due to its high-quality imagery and widely distributed categories, which makes many regions that might otherwise be ambiguous or difficult to annotate explicitly recognizable. Importantly, this characteristic is derived from the annotation of 20 large-scale real RS scenes, rather than from artificially synthesized regions or highly simplified local crops. Therefore, GMSD preserves realistic spatial structures and semantic co-occurrence patterns while reducing the interference of excessive invalid pixels on long-tailed learning. Moreover, the dataset maintains the natural imbalanced distribution. As illustrated in Fig. 5, while head classes account for the vast majority of labeled pixels, tail categories, such as lake and educational area, retain sufficient instances for meaningful long-tailed evaluation.

In summary, the GMSD dataset excels in image quality, annotation granularity, scene representativeness, and the retention of realistic long-tail characteristics, making it an ideal benchmark for long-tailed semantic segmentation in RS.

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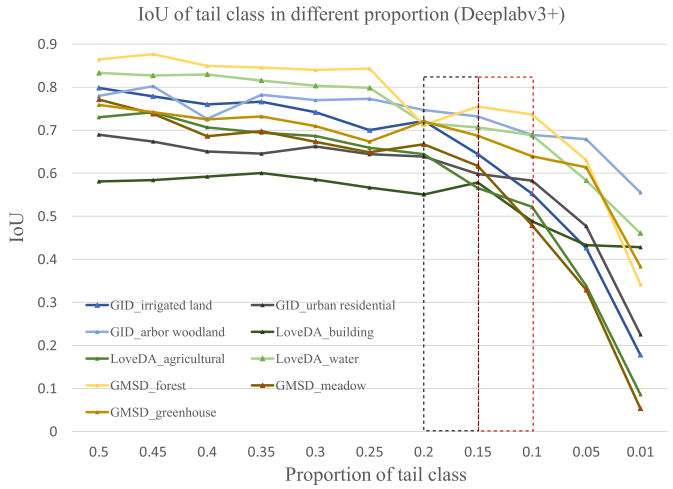


Fig. 3. Visualization of IoU line chart for different land-cover types under varying proportion ratios (τ) based on Deeplabv3 +. The black dashed rectangle highlights a significant declining trend in most tail classes, and the red box indicates that all tail categories suffer from a drastic drop.

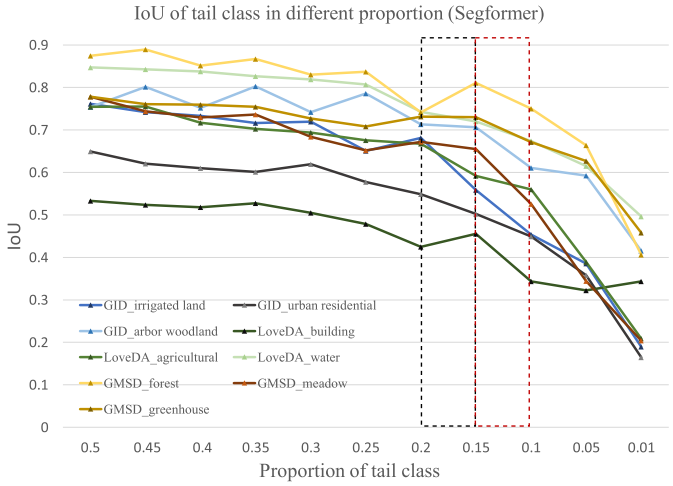


Fig. 4. Visualization of IoU line chart for different land-cover types under varying proportion ratios (τ) based on Segformer. The black dashed rectangle highlights a significant declining trend in most tail classes, and the red box indicates that all tail categories suffer from a drastic drop.

B. Grouping Criteria for Evaluation

In semantic segmentation, labels are assigned pixel-wise, and strong spatial co-occurrence relationships often exist between categories. It is infeasible to directly divide head and tail groups based solely on pixel counts. Most existing segmentation studies rely on personal heuristic rules, leading to inconsistent splits even when using the same dataset, thus reducing reproducibility and lacking rationality.

To address this issue, we propose a novel grouping strategy tailored for semantic segmentation. Specifically, premised on the observation that segmentation performance declines with decreasing tail class proportions in a binary-class problem [42], we seek to find the threshold where tail class accuracy exhibits a sharp decline, indicating significant suppression by head classes. Therefore, we designed a series of binary segmentation experiments where the foreground ratio was uniformly reduced from 0.5 to 0.01, which simulates the learning suppression effect on tail classes. For each target category, a specific dataset is constructed by designating the target as the foreground and the rest as background. Samples were selected to maximize quantity while ensuring the proportion strictly matched the preset ratio. After that, DeepLabv3 + is employed as the baseline with a 7:1:2 data split for each dataset. As shown in Fig. 3, we finally conduct experiments across 11 foreground ratio conditions, allowing for a systematic observation of the varying suppression levels caused by the long-tailed distribution. To ensure the universality and fairness, this validation was performed on nine representative categories in two widely adopted datasets, GID, LoveDA, and our GMSD. Concretely, we select urban residential, arbor woodland, and irrigated land from GID; building, water, and agriculture from LoveDA; forest, meadow, and greenhouse from GMSD.

To further reduce the potential architecture-specific bias in determining the head–tail partition criterion, we additionally repeated the above binary segmentation experiments on both the CNN-based DeepLabv3 + and the Transformer-based SegFormer, under the same experimental settings. As shown in Figs. 3 and 4, the tail-class IoU curves across two frameworks demonstrate highly consistent degradation patterns. When foreground proportion decreases from 50% to 20%, the segmentation accuracy remains relatively stable for most categories. At this stage, the tail class training is subject to considerably less suppression from the head classes, allowing more balanced learning. From 20% to 15%, certain categories experience a sharp decline, such as irrigated land, while others show slow degradation or even minor improvements. Notably, when the proportion decreases from 15% to 10%, a pronounced accuracy drop occurs for all categories, which indicates that head classes exert a substantial suppressive effect on tail classes during training.

Based on these substantial empirical observations, we finally adopt 15% pixel proportion as the grouping threshold for long-tailed semantic segmentation tasks, at which distinct performance disparity occurs. Classes that collectively account for approximately 85% of the dataset are designated as head group, while the remainder are grouped into tail classes. As illustrated in Fig. 5, the proposed criterion yields the head–tail grouping results for the GID, FBP, and GMSD datasets, which is also applied in subsequent experiments to systematically address category imbalance.

C. Adjacency-Filtered Tail Augmentation

Our proposed AFTA module can effectively increase tail class sample diversity and provide more enriched contextual information while preserving geographic plausibility, thus facilitating better feature learning for tail categories. Given the substantial imbalance between head class and tail class instances, only a small number of tail samples in a mini-batch can influence the network optimization. To address this issue, we selectively augment tail samples within each batch based on geographic adjacency priors between different categories.

Initially, prior to training, we statistically analyze the adjacency relationships between classes via a simple

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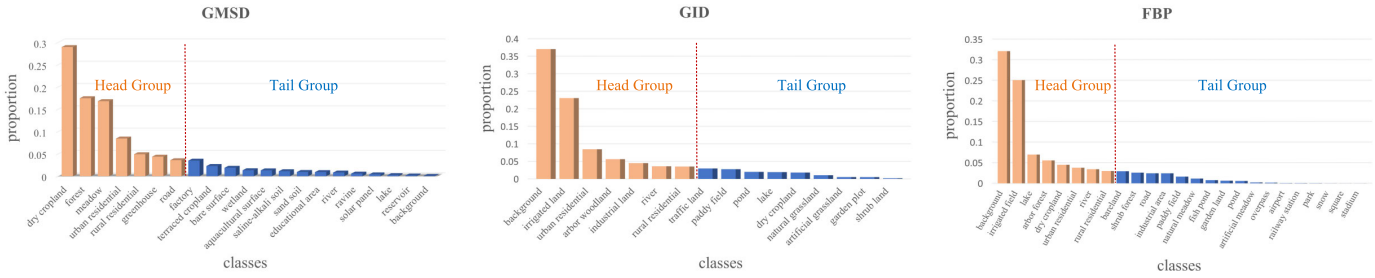


Fig. 5. Category distributions and the division of head–tail groups of the GMSD, GID, and FBP datasets.

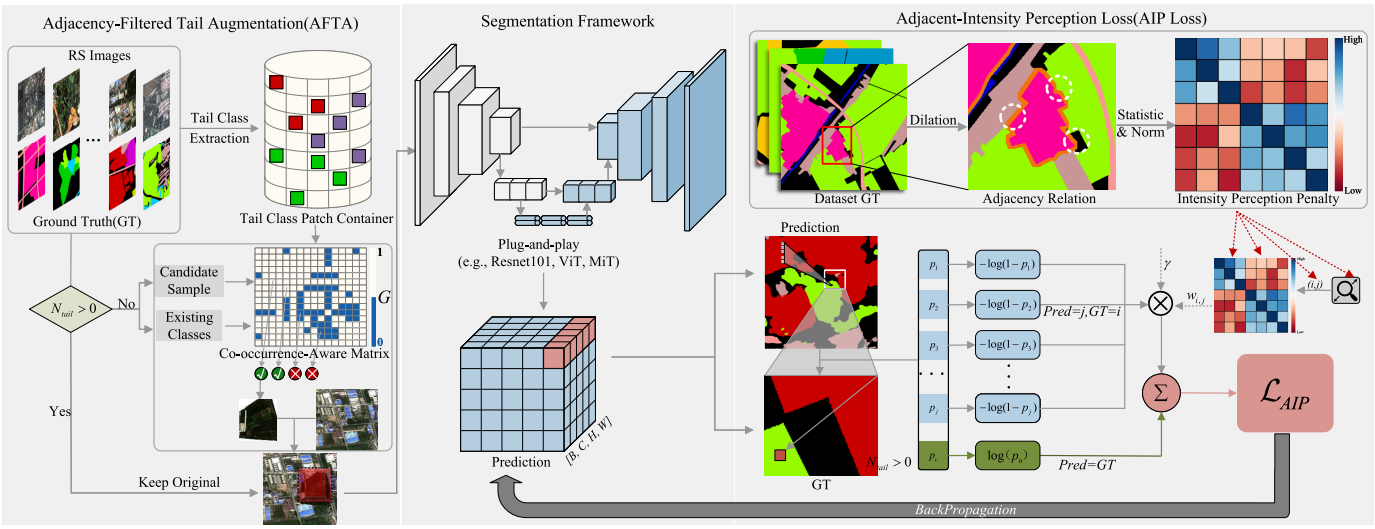


Fig. 6. Framework of our proposed GeoAP, which consists of AFTA module and AIP loss.

morphological dilation operation. This yields the final geographic co-occurrence-aware matrix G, which is subsequently utilized to guide tail-specific sample augmentation. The computation is formulated as follows:

$$G_{ij} = \begin{cases} 1, & if\ \exists k : j \in Y^k_{\chi(r_i)} \\ 0, & otherwise \end{cases} \tag{1}$$

$$\chi(r) = r \oplus B \backslash r \tag{2}$$

each element G_{ij} represents the adjacency relationship between category i and category j in geographic space, where 1 indicates spatially adjacent and 0 implies spatially disjoint, respectively. $Y^k_{\chi(r_i)}$ denotes the label value of the kth sample in region $\chi(r_i)$. And $\chi(r_i)$ is the adjacent area of the region corresponding to class i, where B and $\oplus$ is the structuring element and dilation process, respectively.

An example of such adjacency relationships is G shown in Fig. 6. Head classes typically share spatial adjacency with a wider variety of categories (the last row), while tail classes tend to border fewer classes (the sixth row from the bottom), demonstrating distinct adjacency patterns. This matrix provides valuable cross-category geographic correlation information, which is particularly important in RS imagery interpretation.

During training, for each mini-batch, we first catch all tail regions in the batch according to the tail group defined in Section III-B. These patches are temporarily stored in a dedicated container and used for enhancing the current batch

$$C_{tail} = \{(x_r, y_r, c_r) \,|\, c_r \in S_{tail}\} \tag{3}$$

where C_{tail} represents the temporary container of tail samples. The triplet (x_r, y_r, c_r) denotes the raw image, the label, and the corresponding class of region. Additionally, S_{tail} indicates the set of tail classes in the label.

In order to expand the distribution contexts of the tail class, augmentation is performed for the tail-scarce samples. We first calculate the number of tail regions in each sample. Samples with a count of less than 1 ($g_k = 1$) are selected for the candidate augmentation set, while those with a greater ($g_k = 0$) are kept original. In this way, tail distribution can be effectively improved without compromising the integrity of head samples. The specific process is as follows:

$$N^k_{tail} = \sum_{r \in R} I_{c_r}(S_{tail}) \tag{4}$$

$$g_k = \begin{cases} 1, & N^k_{tail} \le 1 \\ 0, & N^k_{tail} > 1 \end{cases} \tag{5}$$

where N^k_{tail} represents number of tail regions in kth sample. R and I is the all regions in current sample and indicator function, respectively.

For each candidate image, adjacency assessment is performed against the tail regions in the container based on the

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co-occurrence-aware matrix G. As illustrated in Fig. 6, if the existing classes in the candidate image preserve geographic spatial dependency with the selected patch reflected in G, augmentation proceeds; otherwise, another tail patch will be carried. If all patches in the container are geographically incompatible (i.e., mutually exclusive), mixture is skipped for that image

$$V^k = \left\{(x_r, y_r, c_r) \in C_{tail} | G_{ic_r} \neq 0 \quad \forall i \in Y^k\right\} \tag{6}$$

where V^k denotes all selected tail regions that can be blended with the current sample. Y^k is categories included in the label.

Finally, we take one of them and apply mixed augmentation at a random spatial location of the sample for network training

$$X^{AFTA} = X^k \odot (1 - M_{box}) + x_{r^*} \odot M_{box} \tag{7}$$

$$Y^{AFTA} = Y^k \odot (1 - M_{box}) + y_{r^*} \odot M_{box} \tag{8}$$

where X^k and Y^k represent the original image and the label, respectively. $\odot$ denotes pixel-wise multiplication and M_{box} is the randomly selected mixing mask region.

The geographic adjacent filtering process ensures that the augmented samples maintain plausible spatial configurations, avoiding unrealistic situations such as buildings appearing within lakes and greenhouses emerging within wetlands. By explicitly incorporating the inherent spatial and neighboring constraints unique to RS imagery, the AFTA module significantly improves the diversity of tail class distributions while maintaining geographical rationality.

D. AIP Loss

In RS long-tailed semantic segmentation tasks, tail samples often exhibit relatively small spatial scales and marked geographical dependencies. For instance, educational land are frequently surrounded by urban residential areas, whereas grassland is seldom observed nearby. Such interclass spatial correlations constitute the unique geographic knowledge of RS scenes.

Therefore, we propose the AIP Loss, which assists the network to learn geographical dependency strengths between different classes, enhancing their separability. Specifically, similar to the co-occurence-aware matrix G_{ij} mentioned in Section III-C, we further compute the neighboring counts between every pair of categories in the training set in the following equation:

$$A_{ij}^M = \sum_{k=1}^{N} \sum_{r \in R_i^k} I\left(\exists u \in (r \oplus B \backslash r) : Y_u^k = j\right) \tag{9}$$

where A_{ij}^M denotes the spatial neighboring count between class i and class j, N is the total number of training samples, and Y_u^k denotes the class of the adjacent region u of r within the kth label map.

Subsequently, to quantify the perceived adjacency strength between classes, we perform row-wise normalization to construct the transition probability matrix. Unlike conventional global normalization schemes, this process uses the current class as the reference, resulting in a nonsymmetric adjacency intensity matrix. This asymmetry reflects the unique geographical dependency of different land cover categories. For example, even if the raw adjacency count between grass and forest is identical, the normalized adjacency strength can differ because the total adjacency scale of grass is not equal to that of forest. Therefore, we derive distinct interclass perceptual strengths that characterize the specific extent of their adjacency

$$A_{ij}^T = \frac{A_{ij}^M}{\sum_{k=1}^{C} A_{ik}^M} \tag{10}$$

where A_{ij}^T is the normalized adjacency intesity matrix.

After that, global standardization and inversion are performed to derive the final penalty weight based on adjacency strength, making it compatible for training. When two classes are highly dependent in geographical space (i.e., exhibiting high perception strength), segmentation confusion is understandable, given the annotation difficulty and their boundary confusion. To mitigate over-penalization in such cases, we assign a lower loss penalty. Conversely, for pairs of categories that are geographically mutually exclusive, the loss penalty is increased to explicitly suppress these unreasonable predictions

$$A_{ij} = \begin{cases} 1, & if \ \mathrm{i} = \mathrm{j} \\ 1 - \dfrac{A_{ij}^T - \min\left(A_{ij}^T\right)}{\max\left(A_{ij}^T\right) - \min\left(A_{ij}^T\right)}, & if \ \mathrm{i} \neq \mathrm{j} \end{cases} \tag{11}$$

where A_{ij} is the final penalty weight for loss.

When obtaining predictions from the segmentation network, for each pixel result, the entropy loss corresponding to the ground truth is used as basic segmentation guidance without penalty, while the probabilities of nontarget classes are subject to adjacency penalties derived from Matrix A_{ij}. Finally, we get the AIP Loss as follows:

$$\mathcal{L}_{AIP} = -\frac{1}{HW} \sum_{h=1}^{H} \sum_{w=1}^{W} \left(\gamma \cdot \sum_{k \neq y} A_{yk} \log\left(p_k^{(h,w)}\right) + \log\left(p_y^{(h,w)}\right) \right) \tag{12}$$

where H and W denote the image resolution, y denotes the ground-truth label at position (h, w), $p_k^{(h,w)}$ represents the predicted probability of class k for pixel (h, w), A_{yk} signifies the adjacenct penalty weight between label y and class k, and γ is a balancing coefficient used to modulate the scale of the adjacency penalty, default set to 0.1.

We argue that explicitly guiding the model to learn such adjacency relationships can effectively assist the network in improving the discriminative capability for tail classes under complex contextual backgrounds.

To sum up, the AFTA module and the AIP Loss jointly constitute our GeoAP, which allows deeply mine the geographical adjacency relationships among different land covers in RS imagery. This approach effectively assists the network in improving the discriminative capability for tail classes under complex contextual backgrounds. Furthermore, GeoAP is seamlessly integrated into various plug-and-play segmentation networks, demonstrating superior robustness and scalability.

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TABLE II

TEST RESULTS WITH DIFFERENT PLUG-AND-PLAY METHOD ON GMSD, GID, FBP DATASET. BOLD INDICATES THE BEST RESULTS, AND UNDERLINING INDICATES THE SECOND-BEST RESULTS. ALL RESULTS ARE AVERAGED OVER FIVE INDEPENDENT RUNS

Methods	GMSD				GID				FBP			
	OA	mIoU	Head	Tail	OA	mIoU	Head	Tail	OA	mIoU	Head	Tail
Baseline [43]	83.78	61.18	69.34	57.08	77.22	53.93	64.35	45.82	82.08	56.33	75.28	47.41
Focal Loss [15]	83.58	61.04	68.95	57.09	77.62	54.10	64.43	46.06	82.28	56.76	75.33	48.03
RE [44]	83.54	61.42	69.30	57.48	76.78	53.85	63.81	46.10	81.07	55.48	73.97	46.78
WCE [21]	83.38	61.30	69.40	57.35	74.44	53.95	62.31	47.45	80.41	56.98	73.50	49.21
CFL [34]	84.12	61.63	70.00	57.45	77.74	54.62	64.68	46.79	82.40	56.83	75.80	47.91
BLV [35]	83.58	60.79	69.14	56.62	**77.76**	53.91	64.48	45.69	82.48	56.89	75.76	48.01
GeoAP(ours)	**84.64**	**62.80**	**70.88**	**58.76**	77.75	**56.01**	**64.93**	**49.06**	**82.54**	**58.05**	**75.82**	**49.68**

IV. EXPERIMENTS

A. Dataset Description

1) *GMSD Dataset:* The GMSD dataset comprises 20 large-scale scenes, each with a size of 7400 × 8200 pixels, featuring 21 precisely annotated categories. Given that this dataset encompasses various terrestrial scenarios and urban distribution while exhibiting significant class imbalance, it provides excellent learning samples for addressing the problem of realistic long-tailed semantic segmentation. For training purposes, these scenes were cropped into 5760 image patches of size 448 × 448. The dataset was partitioned into training, validation, and testing sets following a classic 7:1:2 ratio, resulting in 4031 images for training, 576 for validation, and 1153 for testing.
2) *GID Dataset:* The GID dataset consists of two subsets: large-scale classification dataset and fine land-cover classification dataset. In this study, we utilized the fine land-cover classification subset, which contains ten scenes of 6800 × 7200 pixels annotated with 16 land-cover categories. The diversity of these categories provides reliable samples for long-tailed feature extraction. We cropped the original scenes into 1820 patches of size 512 × 512. Subsequently, the dataset was divided into 1274 images for training, 182 for validation, and 364 for testing.
3) *FBP Dataset:* The FBP dataset serves as a large-scale semantic segmentation benchmark, containing five billion annotated pixels across 25 land-cover categories. It covers multiple cities and regions nationwide, characterized by substantial variations in land cover distribution, making it an ideal dataset for studying class imbalance. We processed the dataset into 512×512 patches, yielding a total of 27 300 images. These were also allocated for network learning based on a 7:1:2 ratio for training, validation, and testing, respectively.

Fig. 5 illustrates the distribution of different categories across the GMSD, GID, and FBP datasets. Following the criteria described in Section III-B, they are partitioned into corresponding head and tail groups, which are utilized for GeoAP training and the final performance comparison.

B. Overall Performance

Table II presents a comprehensive quantitative comparison between our proposed GeoAP and other state-of-the-art (SOTA) plug-and-play strategies for long-tailed problems, including Focal Loss [15], RE [44], WCE [21], CFL [34], BLV [35], and our GeoAP. Adopting DeepLabv3 + with backbone ResNet101 as the baseline framework, we evaluate performance across four key metrics: overall accuracy (OA), mean IoU (mIoU), and group-wise mIoU for Head and Tail group (categorized based on the criteria in Section III-B). To further reduce the influence of random initialization and training fluctuations, all comparison experiments in Table II are repeated five times with different random seeds. For compactness and readability, we report the averaged results in the main table.

As shown in Table II, real-world RS datasets exhibit evident long-tailed distributions, leading to a substantial performance gap between head and tail classes. For example, on the GID dataset, the baseline shows an mIoU gap of approximately 18.53% between head and tail classes, and similar trends can also be observed on GMSD and FBP. Although existing long-tailed learning strategies can improve the baseline in certain cases, their gains are not always consistent across different datasets and evaluation metrics. In particular, some methods improve tail-class performance at the cost of head-class accuracy, indicating that generic re-balancing strategies may not be sufficiently adapted to complex RS scenarios.

In contrast, GeoAP achieves more stable and balanced improvements across the three datasets. Specifically, GeoAP obtains the best average mIoU on GMSD, GID, and FBP, reaching 62.80%, 56.01%, and 58.05%, respectively. More importantly, GeoAP brings more pronounced improvements for tail classes while maintaining competitive head-class performance. This demonstrates that the proposed geographical adjacency modeling can effectively alleviate the learning bias toward dominant head classes and strengthen the representation of underrepresented tail categories.

To further validate the generalization capability of our GeoAP, Table III presents an extensive comparative evaluation across a diverse spectrum of segmentation architectures, including Deeplabv3 + [43], Segformer [45], FactSeg [46], RSSformer [47], FarSegPP [48], Mask2former [49], FM [26],

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TABLE III

GENERALIZATION PERFORMANCE OF OUR GEOAP METHOD ACROSS DIFFERENT SEGMENTATION NETWORKS EVALUATED ON GID, GMSD, AND FBP DATASETS. * DENOTES THE FRAMEWORK IS DESIGNED FOR LONG-TAILED SEMANTIC SEGMENTATION. AFTA INDICATES THAT THE NETWORK CANNOT UTILIZE THE COMPLETE GEOAP AND EMPLOYS ONLY THE AFTA MODULE

Method	GMSD				GID				FBP			
	OA	mIoU	Head	Tail	OA	mIoU	Head	Tail	OA	mIoU	Head	Tail
Deeplabv3+ [43]	83.54	61.04	69.08	57.03	77.13	53.72	64.11	45.64	81.97	56.41	75.16	47.58
+GeoAP(ours)	**84.28**	**62.5**	**70.41**	**58.55**	**77.80**	**56.37**	**65.13**	**49.56**	**82.40**	**57.01**	**75.60**	**48.25**
Segformer [45]	87.28	68.81	75.59	65.42	**82.08**	61.85	**69.78**	55.68	83.33	59.25	76.62	51.08
+GeoAP(ours)	**87.75**	**69.67**	**76.13**	**66.44**	81.99	**62.2**	69.50	**56.52**	**83.48**	**59.84**	**76.87**	**51.83**
FactSeg [46]	83.32	61.17	**69.86**	56.83	77.44	55.47	65.05	48.02	81.28	55.65	68.69	50.57
+GeoAP(ours)	**83.99**	**61.61**	69.70	**57.56**	**79.64**	**59.43**	**67.42**	**53.21**	**82.36**	**57.49**	**70.04**	**52.62**
RSSFormer [47]	83.9	60.96	65.84	58.52	75.79	54.4	62.22	48.31	81.76	54.30	74.45	44.82
+GeoAP(ours)	**85.76**	**64.37**	**69.50**	**61.81**	**78.23**	**56.12**	**64.92**	**49.27**	**82.13**	**54.99**	**75.14**	**45.51**
FarSegPP [48]	86.61	66.91	74.51	63.11	81.25	59.58	**69.61**	51.79	83.18	58.98	76.06	50.94
+GeoAP(ours)	**87.79**	**69.5**	**76.83**	**65.83**	**81.42**	**60.54**	69.28	**53.74**	**83.59**	**59.85**	**77.26**	**51.66**
Mask2Former [49]	84.88	62.92	72.66	58.04	77.77	53.7	64.74	45.11	81.4	55.86	**74.18**	47.24
+AFTA(ours)	**85.73**	**65.13**	**74.08**	**60.65**	**78.85**	**55.94**	**65.35**	**48.62**	**81.56**	**56.44**	74.09	**48.13**
FM [26]	85.41	64.89	73.60	60.54	78.57	**55.56**	65.69	47.69	81.31	56.04	73.82	47.67
+AFTA(ours)	**85.39**	**65.44**	**73.70**	**61.32**	**79.57**	56.51	**66.92**	**48.41**	**81.57**	**56.47**	**74.06**	**48.19**
GDGNet [50]	81.38	61.24	67.48	53.74	78.26	52.62	**65.33**	43.67	82.65	53.12	75.42	43.29
+GeoAP(ours)	**82.17**	**63.59**	**68.13**	**56.78**	**78.48**	**53.23**	65.28	**44.78**	**83.51**	**54.74**	**76.27**	**45.27**
MSEONet [51]	86.46	67.06	74.14	63.52	79.39	57.19	67.26	49.36	83.29	57.44	76.41	48.25
+GeoAP(ours)	**87.03**	**68.44**	**74.52**	**65.40**	**80.21**	**59.31**	**68.02**	**52.54**	**83.34**	**58.44**	**76.87**	**49.76**

GDGNet [50], and MSEONet [51]. We assess the adaptability of GeoAP by integrating it into the latest deep learning frameworks, ranging from standard semantic segmentation backbones (e.g., DeepLabv3 +, SegFormer) to specialized architectures designed for long-tailed distributions (e.g., FactSeg, RSSFormer).

For each baseline, we incorporated the proposed GeoAP module and benchmarked it against the vanilla model. It is worth noting that for specific architectures. such as Mask2Former and FM, which are structurally incompatible with the full GeoAP framework, we only deployed the AFTA module instead. The comprehensive experiments across three datasets demonstrate that GeoAP consistently enhances performance in both head and tail classes, regardless of the underlying network architecture. This highlights the module's exceptional interoperability, suggesting that our approach offers complementary benefits to current SOTA solutions.

C. Comparison on the GMSD Dataset

In this section, we present a comprehensive comparison between the proposed GeoAP and various plug-and-play modules on the GMSD dataset. Furthermore, we investigate the generalization capability of GeoAP across different segmentation frameworks to demonstrate its robustness in practical applications.

1) Performance Comparison With Long-Tailed Methods: Table II details the quantitative results of various approaches on the GMSD dataset. Benefiting from the AFTA module and AIP loss, we have achieved the highest mIoU of 62.80% and a tail-group mIoU of 58.76%, substantially improving tail-class performance while maintaining the accuracy of head classes.

Fig. 7 visualizes the relative performance gains of different methods compared to the baseline for each land-cover category (excluding background) on the GMSD dataset. Specifically, we normalize the results by setting the baseline accuracy to 1.0; the scores of other methods are then calculated as ratios relative to this baseline. The labels C1–C20 are color-coded, where blue and red denote the head and tail groups, respectively. It is evident that our method achieves the highest recognition accuracy across the vast majority of categories. Notably, the improvements on tail classes, such as C8, C9, and C16, are remarkably superior to other methods. Although slight performance dips are observed in C14 and C18, such minor fluctuations are inevitable in complex multiclass datasets. Overall, our method demonstrates a robust capability to deliver a more balanced performance improvement for long-tailed semantic segmentation.

To visually demonstrate the improvements on tail classes, Fig. 8 presents part of the samples from tail categories across different methods. Benefiting from the GeoAP's effective exploitation of adjacency relationships, our method yields superior recognition accuracy and boundary delineation for tail classes. As observed in the first row regarding Wetland, most competing methods misclassify these regions as Meadow due to the high feature similarity between them. In contrast, by integrating contextual information from the adjacent River, GeoAP effectively assists the network in rectifying errors in these challenging, ambiguous regions. Similar observations apply to the Ravine in the fifth row and the River in the eighth

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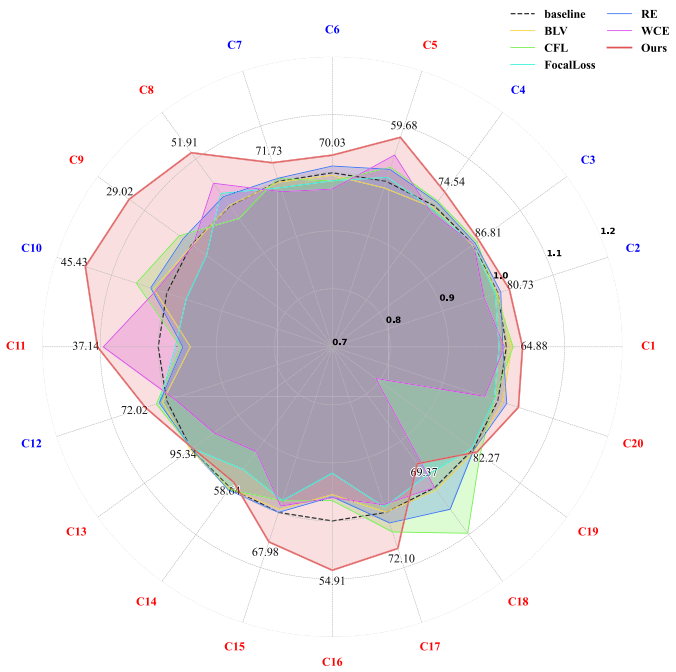


Fig. 7. Performance comparison of different methods on the GMSD dataset. The black dashed line denotes the baseline, the red line shows the relative improvement of GeoAP, and values are the IoU (%). Blue and red denote the head and tail classes, respectively, (C1) terraced cropland. (C2) dry cropland. (C3) forest. (C4) meadow. (C5) wetland. (C6) rural residential. (C7) urban residential. (C8) factory. (C9) educational area. (C10) road. (C11) bare surface. (C12) greenhouse. (C13) solar panel. (C14) ravine. (C15) saline-alkali soil. (C16) sand soil. (C17) river. (C18) lake. (C19) reservoir. (C20) aquacultural surface.

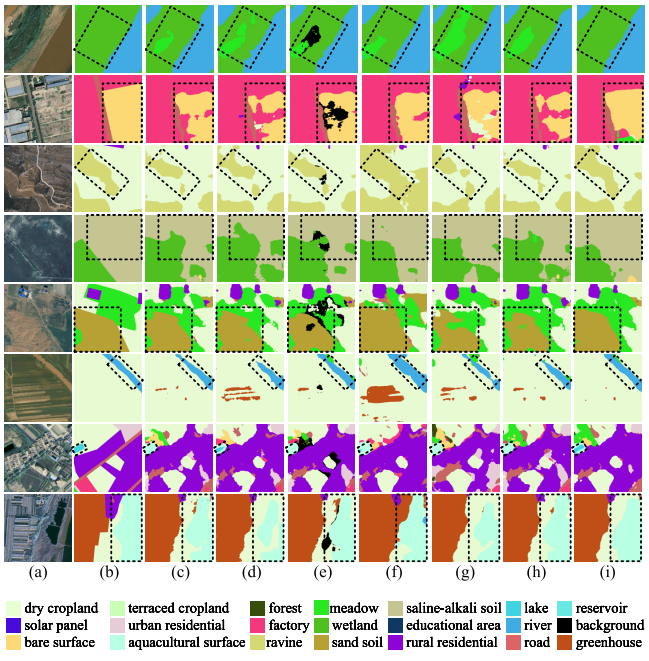


Fig. 8. Visual comparison of different methods on tail class samples on GMSD dataset. (a) Original image. (b) Label. (c) Baseline. (d) Focal loss. (e) RE. (f) WCE. (g) CFL. (h) BLV. (i) GeoAP(ours).

row, where the utilization of adjacency cues significantly contributes to more precise segmentation for tail categories.

2) Generalization Improvements Across Different Segmentation Frameworks: Table III presents a comprehensive

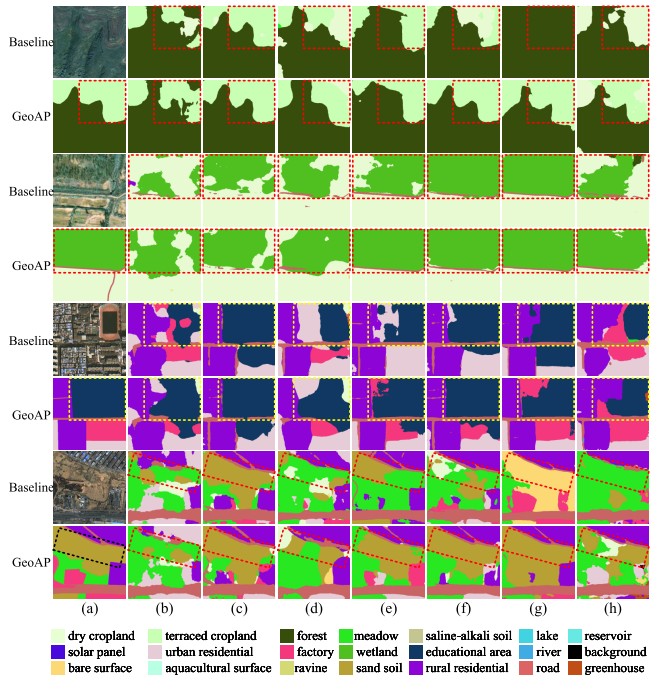


Fig. 9. Generalization improvements of GeoAP across different segmentation networks on GMSD dataset. (a). Original image/label. (b) Deeplabv3+. (c) Segformer. (d) FactSeg. (e) Mask2Former. (f) FarSegPP. (g) FM. (h) RSSFormer.

performance improvement of various segmentation frameworks on the GMSD dataset, both with and without the integration of our GeoAP. As illustrated, our approach consistently delivers significant improvements on all metrics for every framework, notably yielding a tail-class gain of more than 1% in most cases. Moreover, even for architectures like Mask2Former and FM, where only the AFTA module is applicable due to structural constraints, they still exhibit remarkable performance improvements. Therefore, the integration of GeoAP effectively demonstrates its strong adaptability and robustness in addressing long-tailed semantic segmentation challenges.

Fig. 9 visualizes the improvements on some tail class samples across different networks. As illustrated in the second sample, where severe confusion exists between Wetland and Dry Land, the GeoAP module achieves more precise region extraction by effectively leveraging intrinsic physical characteristics and mining the adjacency correlations with the surrounding dry land. Furthermore, in the third sample, buildings within educational areas exhibit extremely high visual similarity to those in residential areas, posing a challenge for baseline models. Benefiting from the effective enrichment of tail class distribution scenarios in TACA, our method achieves superior recognition accuracy, effectively distinguishing between these semantically confusing categories.

D. Comparison on the GID Dataset

1) Performance Comparison With Long-Tailed Methods: We conducted a comprehensive quantitative comparison and analysis on the GID dataset reported in Table II. It is evident that GeoAP achieves the SOTA holistic performance, yielding

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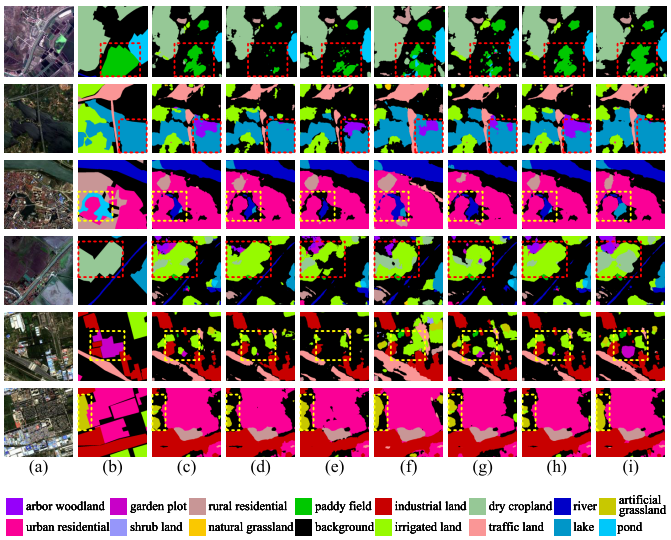


Fig. 10. Visual comparison of different methods on tail class samples on GID dataset. (a) Original image. (b) Label. (c) Baseline. (d) Focal loss. (e) RE. (f) WCE. (g) CFL. (h) BLV. (i) GeoAP(ours).

an mIoU of 56.01% and tail mIoU of 49.06%. Compared to other long-tailed approaches, GeoAP demonstrates superior stability and balance. Specifically, while WCE attains a high score on tail classes (47.45%), it suffers from a significant performance degradation on head classes (dropping to 62.31%). This indicates that naive re-weighting strategies tend to over-suppress head categories. In contrast, attributed to the AFTA module, which diversifies the background distribution of tail classes within adjacency relations, and the AIP loss, which penalizes boundary misclassifications, GeoAP achieves competitive performance on tail classes while securing the highest accuracy on head classes among all compared methods.

To provide intuitive insights, the Fig. 10 visualizes the segmentation results for typical tail categories, such as Paddy field and Lake. As observed in the Paddy field and Lake, baseline and competing methods often produce fragmented regions or blurred boundaries due to sample scarcity and texture confusion. Conversely, the segmentation maps generated by our method exhibit significantly better structural completeness and boundary delineation. Moreover, in the Garden plot scenario (row 5), where other methods struggle with confusion between vegetation and background, GeoAP successfully rectifies these ambiguities. This improvement is largely credited to the AIP module, which leverages adjacency constraints to guide precise pixel-level classification. These visualization results strongly corroborate the advanced nature of our GeoAP for addressing long-tailed semantic segmentation in complex RS scenes.

2) Generalization Improvements Across Different Segmentation Frameworks: To comprehensively evaluate the universality and robustness of the proposed method, we benchmarked the performance of various segmentation frameworks before and after integrating GeoAP. The quantitative results in Table III demonstrate the superior "plug-and-play" capability of GeoAP. Consistent performance gains are observed across all tested baselines, regardless of whether the underlying architecture is CNN-based (DeepLabv3 +) or Transformer-based (SegFormer), or even specifically designed for class

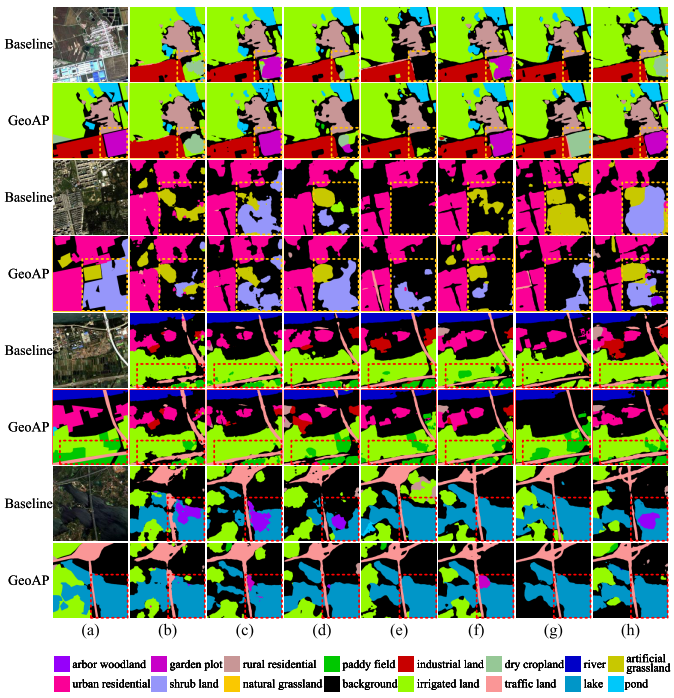


Fig. 11. Generalization improvements of GeoAP across different segmentation networks on GID dataset. (a) Original image/label. (b) Deeplabv3 +. (c) Segformer. (d) FactSeg. (e) Mask2Former. (f) FarSegPP. (g) FM. (h) RSSFormer.

imbalance. This suggests that the proposed AFTA and AIP module can address challenges of long-tailed semantic segmentation that architectural modifications alone cannot fully resolve. Consequently, GeoAP acts as an orthogonal enhancement module, seamlessly integrating into diverse segmentation paradigms to bolster discriminative power for tail classes.

Fig. 11 provides a visual comparison of segmentation results across different baselines. Observing columns (b) through (h), it is evident that while different networks exhibit varying degrees of failure on tail classes (e.g., fragmented predictions in the paddy field in the third sample, or blurred boundaries near water bodies in the fourth sample), the integration of GeoAP yields a consistent and robust improvement across all frameworks. Through explicitly modeling adjacency relationships, GeoAP effectively rectifies these errors, producing semantic masks that are spatially continuous and internally coherent.

E. Comparison on the FBP Dataset

1) Performance Comparison With Long-Tailed Methods: To intuitively evaluate the effectiveness of the proposed GeoAP, we conducted extensive experiments against other SOTA methods on the FBP dataset. As shown in Table II, our method demonstrates superior overall performance, achieving the highest mIoU of 58.05% and outperforming all competing methods. In the challenging tail categories, while WCE shows a high tail score, it suffers a catastrophic drop in OA and head class performance. Conversely, our GeoAP realizes a balanced performance gain while substantially improving tail class accuracy. This indicates that our proposed AFTA and AIP modules fully consider the intrinsic geographic attributes

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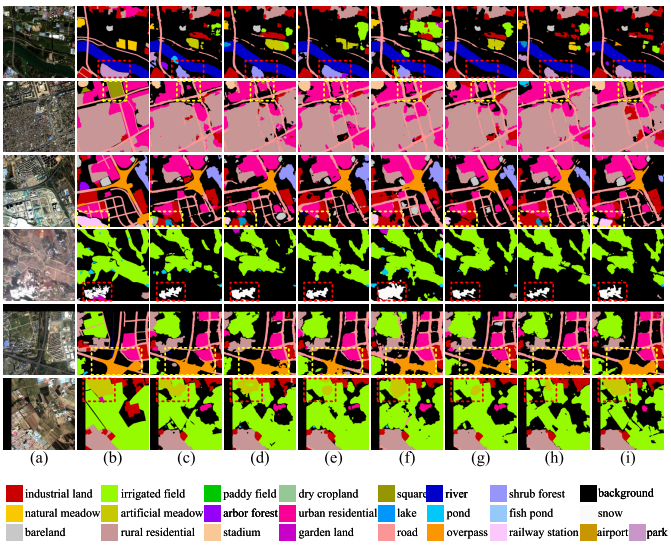


Fig. 12. Visual comparison of different methods on tail class samples on FBP dataset. (a) Original image. (b) Label. (c) Baseline. (d) Focal loss. (e) RE. (f) WCE. (g) CFL. (h) BLV. (i) GeoAP(ours).

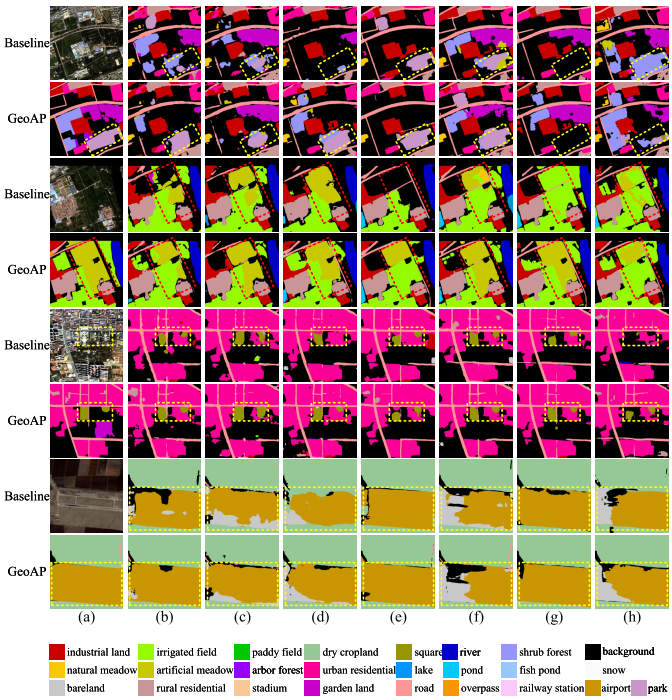


Fig. 13. Generalization improvements of GeoAP across different segmentation networks on FBP dataset. (a). Original image/label. (b) Deeplabv3 +. (c) Segformer. (d) FactSeg. (e) Mask2Former. (f) FarSegPP. (g) FM. (h) RSSFormer.

of both head and tail classes while enhancing tail class representations, thereby achieving the optimal tradeoff compared to other methods.

Fig. 12 visualizes the qualitative segmentation results, which clearly show that the competing methods frequently struggle with misclassification and boundary ambiguity in tail scenarios. For instance, in the "snow" row, most methods fail to detect the small, scattered snow regions or produce fragmented masks. In contrast, GeoAP accurately captures the complete snow region and closely matches the ground truth. This improvement is largely attributed to the AFTA module, which effectively augments the scene distribution of rare classes, preventing the model from overlooking them due to data scarcity. Furthermore, in complex scenes such as "railway station" and "artificial meadow," other methods often confuse the target class with qualitative perspectives. The results demonstrate that the application of GeoAP consistently yields stable performance improvements, with a particularly significant increase in tail class accuracy.

2) Generalization Improvements Across Different Segmentation Frameworks: Table II and Fig. 13 report the performance comparison of six representative segmentation methods with and without the proposed GeoAP, validating the universality and effectiveness from both quantitative and qualitative perspectives. The results demonstrate that the application of GeoAP consistently yields stable performance improvements, with a particularly significant increase in tail class accuracy.

As observed in the visual samples, baseline methods often fail to detect large contiguous regions of tail classes, resulting in severe "holes" or complete misclassification, including the artificial meadow and airport. In contrast, the methods equipped with GeoAP successfully recover these missing regions. Moreover, in complex scenes such as a park, the GeoAP-enhanced models produce significantly smoother and more semantically coherent masks by introducing the AIP loss, which penalizes unreasonable interclass adjacencies. These

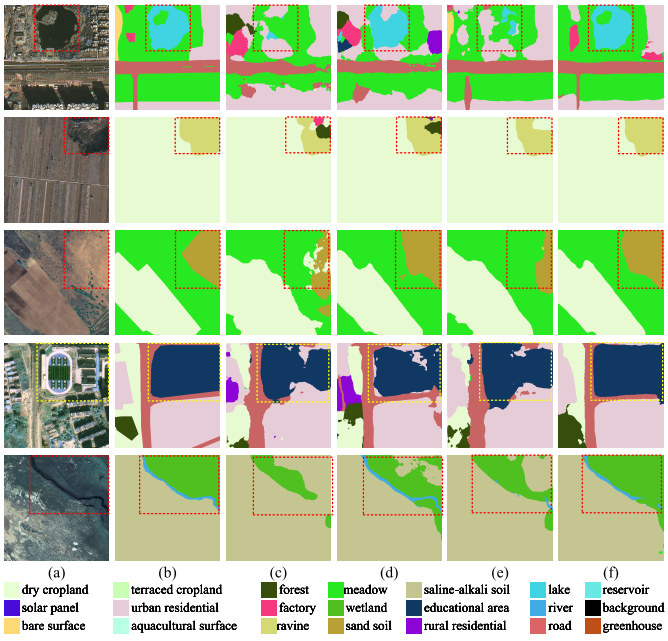


Fig. 14. Visual comparison of different modules in the ablation study of GeoAP on the GMSD dataset. (a) Image. (b) Label. (c) Baseline. (d) w/AFTA. (e) w/AIP loss. (f) w/GeoAP.

results demonstrate that our method achieves robust feature learning and structural integrity of the semantic scenes.

F. Ablation Study

To systematically investigate the contribution of each component within our proposed GeoAP, we conduct ablation studies on the GMSD dataset. Using DeepLabv3 + as the

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TABLE IV
ABLATION STUDY OF THE PROPOSED GEOAP, BOLD INDICATES THE BEST RESULTS

Model	OA	mIoU	Head	Tail
Baseline	83.54	61.04	69.08	57.03
Baseline+AFTA	84.18	61.91	69.95	57.9
Baseline+AIP	84.18	62.07	70.40	57.91
Baseline+GeoAP	**84.28**	**62.50**	**70.41**	**58.55**

baseline, we incrementally incorporate the AFTA module and the AIP Loss. Table IV and Fig. 14 present the quantitative performance comparisons and visual improvements, respectively.

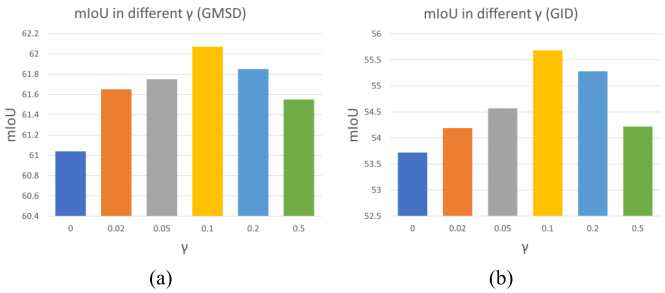
As shown in Table IV, we observe a clear performance improvement for tail categories when integrating the AFTA module. This success verifies that AFTA effectively mitigates the scarcity of rare category samples. Unlike standard augmentation, AFTA enriches the diversity of tail classes within topologically valid contexts, preventing the model from overfitting to limited tail samples while maintaining semantic consistency. When the AIP Loss is applied, a substantial boost in head class also occurs, which suggests that explicitly modeling adjacency intensity benefits the global structural understanding of the scene. Through penalizing violations of neighbor relationships, it refines the boundaries for both dominant and rare classes, leading to a more robust representation of the overall layout.

When combining both AFTA and AIP Loss, GeoAP achieves the SOTA performance and the remarkable improvement of tail class accuracy. This demonstrates that the two modules are complementary: AFTA provides the necessary data diversity to wake up the network's attention to rare classes, while AIP Loss provides strong geometric constraints to ensure precise boundaries.

The visualization results in Fig. 14 highlight the model's capability to handle semantic ambiguity and boundary erosion in complex tail scenarios. Taking the second row as an example, the ravine region occupies only a small portion of the image and is surrounded by large-area dry cropland, which makes it easily suppressed under the long-tailed distribution. The baseline fails to accurately recognize this tail category and produces incomplete or unstable predictions in the highlighted region. After introducing AFTA, the model better preserves the spatial extent and structural completeness of ravine regions. With AIP Loss, the prediction results show cleaner boundaries and fewer unreasonable category confusions. When AFTA and AIP Loss are jointly used, GeoAP produces the result closest to the ground truth, verifying the complementary effectiveness of the two modules. Similar improvements are consistently observed in other minority classes, including lakes and educational areas, validating its effectiveness in handling long-tailed semantic segmentation.

G. Sensitivity Analysis

To investigate the influence of the balancing coefficient γ in the AIP Loss, we conduct additional experiments for hyperparameter sensitivity analysis. Specifically, AIP Loss is incorporated into DeepLabv3 + with different γ values selected from 0, 0.02, 0.05, 0.1, 0.2, and 0.5, so as to evaluate the effect of varying adjacency-intensity perception strengths.



(a) (b)

Fig. 15. Sensitivity analysis of the hyperparameter γ in terms of mIoU on two datasets. (a) Results on the GMSD dataset. (b) Results on the GID dataset.

TABLE V
COMPLEXITY ANALYSIS OF GEOAP ON DEEPLABV3 + AND SEGFORMER. COMPUTATIONAL COST (FLOPS) IS MEASURED AT THE COMPLETE TRAINING-STEP LEVEL, INCLUDING DATA PROCESSING, FORWARD PROPAGATION, LOSS COMPUTATION, AND BACKWARD PROPAGATION

Model	Size	Params	FLOPs	Train (epoch)	FPS	Memory
Deeplabv3+	512^2	58.75M	112.75G	0.59min	73	1.03GB
+GeoAP	512^2	58.75M	112.76G	0.7min	73	1.03GB
Segformer	512^2	27.36M	198.31G	0.37min	108	1.28GB
+GeoAP	512^2	27.36M	198.33G	0.45min	108	1.28GB

As shown in Fig. 15, GeoAP achieves stable performance across a moderate range of γ. Compared with $\gamma = 0$, the introduction of the adjacency-intensity perception term consistently improves the mIoU on both GMSD and GID datasets, demonstrating the effectiveness of geographical adjacency guidance. The best performance is obtained when $\gamma = 0.1$, indicating that a proper adjacency penalty can help the model suppress geographically implausible predictions while maintaining normal semantic discrimination.

When γ is further increased to 0.5, the mIoU decreases slightly, suggesting that an excessively large adjacency penalty may over-constrain the optimization process. Therefore, we set $\gamma = 0.1$ as the default value in all experiments. The sensitivity analysis confirms that GeoAP is robust to the choice of γ and can consistently improve long-tailed semantic segmentation performance across different datasets.

H. Complexity Analysis

To better substantiate the plug-and-play and scalable properties of GeoAP, we added complexity analysis in Table V, including parameter count, step-level computational cost, training time, inference speed, and GPU memory consumption. All experiments are conducted on a single NVIDIA 5090 GPU with 32 GB of RAM and the training time statistics are based on the GID dataset.

Specifically, we evaluate GeoAP on two representative segmentation frameworks, i.e., DeepLabv3 + and SegFormer. The results show that introducing GeoAP does not increase the

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number of learnable parameters since it does not introduce additional learnable layers or inference branches into the segmentation backbone. In addition, GeoAP only brings a negligible increase in step-level FLOPs. For DeepLabv3 +, the computational cost increases from 112.75 to 112.76 G, while for SegFormer, it increases from 198.31 to 198.33 G.

We also observe that the training time increases moderately after introducing GeoAP. This additional training cost mainly comes from the AFTA module, which performs tail-region extraction, adjacency filtering, and patch-level augmentation within each mini-batch. Different from pure matrix-based operations, these procedures slightly increase training time. Nevertheless, this overhead only appears during training and remains acceptable considering the consistent performance improvements on tail classes.

More importantly, GeoAP introduces no additional GPU memory consumption and inference burden. The inference FPS remains unchanged for both DeepLabv3 + and SegFormer, i.e., 73 and 108 FPS, respectively.

Overall, the above analysis indicates that GeoAP mainly introduces a limited training-stage overhead, while keeping the deployed inference model unchanged. Therefore, GeoAP provides an effective and deployment-friendly solution for RS long-tailed semantic segmentation.

V. Conclusion

In this work, we focus on addressing the challenging problem of long-tailed semantic segmentation in RS. Given the lack of sufficient exploration in this field, we first construct a new large-scale, multiscene benchmark, aiming to alleviate the data scarcity. A criterion for head–tail partitioning is also established to standardize evaluation for future research. After that, to address the performance degradation influenced by long-tailed distribution, we present GeoAP, a novel framework grounded in the spatial adjacency properties of ground objects. This method expands the tail class to various scenes and explicitly guides the network to learn interclass boundaries, thereby achieving balanced segmentation and demonstrating superior generalization capabilities.

However, the 15% threshold for partitioning is more inclined to be an empirical and task-oriented criterion for RS long-tailed semantic segmentation, where dense pixel-wise prediction and strong spatial co-occurrence make instance-count-based partitioning unsuitable. In our work, this threshold is supported by consistent observations across different datasets, categories, and backbone architectures. It provides a practical boundary at which tail classes begin to be substantially suppressed by dominant classes, thereby enabling more consistent evaluation and targeted optimization. For datasets with substantially different category systems, spatial resolutions, or annotation protocols, the threshold may be different but the same proportion-controlled validation strategy can be adopted.

In addition, it should also be noted that the current GeoAP mainly focuses on boundary-level geographic adjacency. Although the binary adjacency used in AFTA provides a robust and effective plausibility constraint for tail-class augmentation, it does not explicitly represent more complex spatial patterns, such as directional adjacency and enclosure topology. Therefore, richer spatial pattern modeling remains a limitation of the current framework and a promising direction for future work. We will further investigate direction-aware, scale-aware, and topology-aware geographic priors to enhance the representation of complex land-cover relationships in long-tailed RS semantic segmentation. Since GeoAP does not explicitly leverage network architecture to capture RS characteristics, designing specialized architectures tailored to these features provides another feasible direction.

To sum up, we hope that by introducing our GMSD benchmark and GeoAP framework, this work offers fresh perspectives to inspire future research in long-tailed semantic segmentation.

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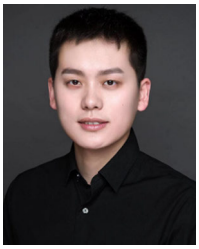
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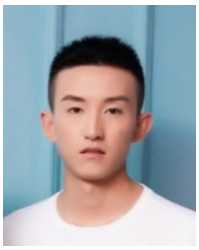
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